

Optics and visual transduction

Focus the image, then convert photons into a retinal signal

Refraction explain the blur

Two surfaces do the focusing

The cornea supplies about two thirds of the eye's refractive power, mainly at the air-cornea interface. The lens supplies about one third and can change power during accommodation. In emmetropia, parallel rays from a distant target focus on the retina while accommodation is relaxed. Refraction depends on both optical power and globe length, so identify where the image would focus before choosing a corrective lens.

Focus position predicts the correction

Error	Mechanism / focus	Correction
Myopia	Long globe or excessive power; distant rays focus in front of retina	Concave, minus lens
Hypermetropia	Short globe or insufficient power; distant rays focus behind retina	Convex, plus lens
Astigmatism	Unequal power between meridians; no single sharp focus	Cylindrical / toric correction
Presbyopia	Age-related loss of lens accommodation; near focus fails	Plus addition for near work

Use the clinical pattern

- Distance blur with relatively clear near vision suggests myopia. Young hypermetropes may compensate by accommodating, so distance vision can remain clear.
- Presbyopia develops despite a working ciliary muscle because the ageing lens becomes less flexible.
- Astigmatism can blur near and distance vision; changing the optical meridian matters.

Cataract scatters light through lens opacity rather than shifting the focus position.

Accommodation muscle contracts, zonules relax

Switching from distance to near

- Near target produces more divergent incoming rays
- ↓
- Parasympathetic activity through CN III increases
- ↓
- Ciliary muscle contracts; ciliary ring diameter decreases
- ↓
- Zonular tension falls, allowing the elastic lens to round up
- ↓
- Central lens thickness and refractive power increase
- ↓
- Near image comes into focus on the retina

The near triad

Response	Effector	Purpose
Accommodation	Ciliary muscle, zonules and lens	Adds focusing power
Convergence	Both medial recti via CN III	Aligns the visual axes
Miosis	Sphincter pupillae via parasympathetic CN III	Improves depth of focus

Predict the opposite state

- For distance viewing, the ciliary muscle relaxes, zonular tension increases and the lens flattens. Lens power falls.
- Zonules are passive suspensory fibres. They do not actively contract to make the lens thicker.
- Parasympathetic blockade impairs accommodation and pupillary constriction; the patient may develop near blur and a larger pupil.
- In presbyopia, extra near power from reading glasses substitutes for declining lens flexibility.

Pinhole is a useful clue

A pinhole restricts peripheral rays and increases depth of focus. A substantial improvement in distance acuity supports an uncorrected refractive component. Failure to improve should prompt consideration of media, retinal or optic nerve pathology; it does not establish one specific diagnosis. Test each eye separately because the better eye can conceal unilateral loss.

Rods, cones and the meaning of visual loss

Match the receptor to the task

Feature	Rods	Cones
Best setting	Dim light, scotopic vision	Bright light, photopic vision
Sensitivity / detail	High sensitivity; lower spatial detail	Lower sensitivity; high spatial detail
Colour	One pigment system; no colour discrimination	S, M and L cone pigments support colour
Distribution	Numerous away from the foveal centre	Highest density at the foveal centre
Typical deficit	Night blindness, poor dim-light navigation	Reduced fine acuity and colour discrimination

Landmarks that change function

- The foveal centre has tightly packed cones and displaced overlying retinal layers, supporting fine acuity. The central foveola is rod-free; the whole macula is not.
 - The optic disc contains no photoreceptors and produces the physiological blind spot in the temporal visual field.
 - Central loss impairs reading and face recognition.
- Extensive peripheral loss impairs navigation, even when Snellen acuity remains good.

Colour is a comparison between cones

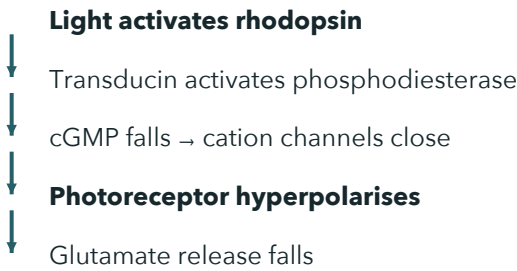
Overlapping S, M and L sensitivities let the brain compare cone responses. Deuteranomaly alters the M-cone pigment response and causes red-green confusion. Deuteranopia lacks functional M-cone input; anomaly and absence are different mechanisms. Ishihara plates screen red-green deficiency, rather than every form of colour dysfunction.

Apply the distribution

A faint star may be easier to see when looking slightly beside it: its image then reaches rod-rich retina. Direct fixation places it on the less light-sensitive foveal cone system.

Phototransduction light reduces the dark current

From photon to signal



Dark and light states

Variable	Dark	Light
cGMP-gated cation channels	Open	Close as cGMP falls
Na+ / Ca2+ inward current	Dark current persists	Inward current decreases
Photoreceptor potential	Relatively depolarised	Hyperpolarised
Glutamate release	Continuous, relatively high	Reduced

The chemical trigger

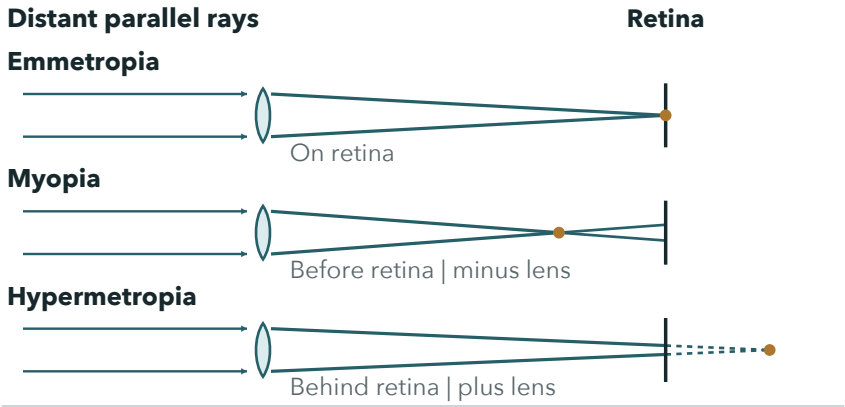
Rhodopsin contains opsin and vitamin A-derived 11-cis-retinal. Light converts retinal to all-trans. As inward cation current falls, continuing outward K⁺ current hyperpolarises the cell. Phosphodiesterase inhibition opposes the light response.

Signal relay and recovery

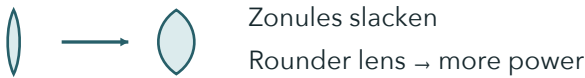
- Photoreceptors use graded changes in membrane potential. Retinal ganglion cells generate the action potentials carried by the optic nerve.
- Reduced glutamate release depolarises ON bipolar cells and hyperpolarises OFF bipolar cells, allowing parallel encoding of light increments and decrements.
- The retinal pigment epithelium supports chromophore recycling and removes shed outer-segment discs. Vitamin A deficiency can impair pigment regeneration, particularly dim-light vision.
- Dark adaptation takes time after bright-light exposure as sensitivity recovers; rods provide the greater eventual sensitivity.

Optical atlas and signal states

A Focus position and near power

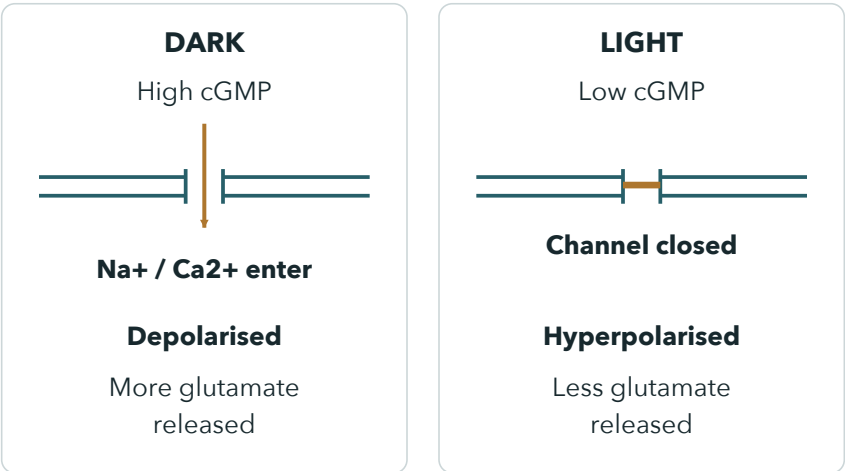


NEAR: ciliary muscle contracts



Distant rays shown with accommodation relaxed. Combined eye power is schematic; dashed rays extrapolate a focus behind the retina.

B The channel switch



Light closes cGMP-gated channels. Photoreceptors hyperpolarise and release less glutamate; ganglion cells supply the optic-nerve spikes.

Reference map

A: TBL 2 slides 14-20; NEI refractive errors. B: slides 28-31; NIH Neuroscience phototransduction. Original schematics.